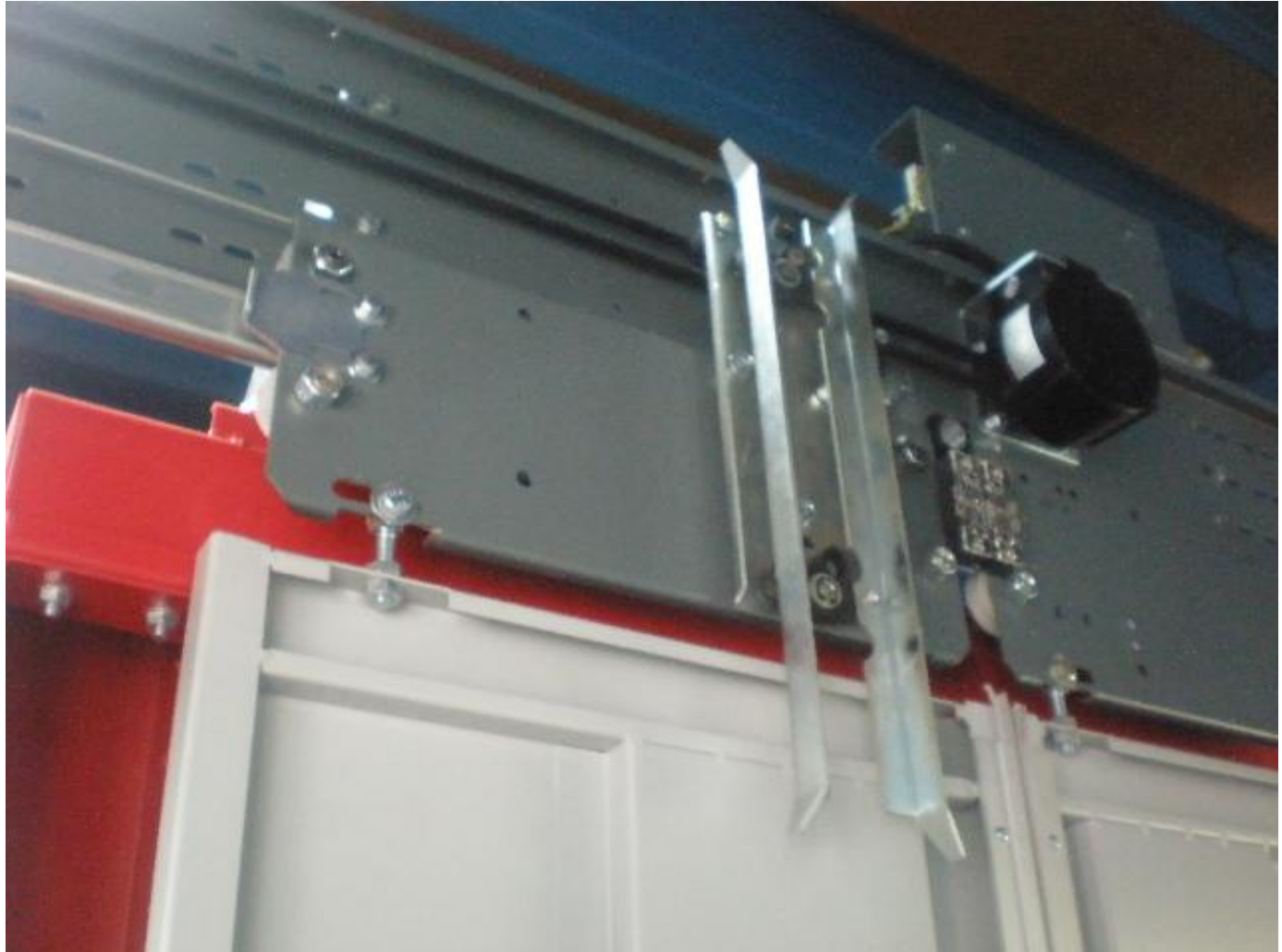


PM Door Operator Training

| PM Door Operator Presentation
1.The front



The Rear



2.PM door operator opening width

Central Opening application:

600 700 750 800 850 900 950 1000 1050 1100

Side opening application:

600 700 750 800 850 900 1000 1100 1200 1300

3.PM Motor & Controller

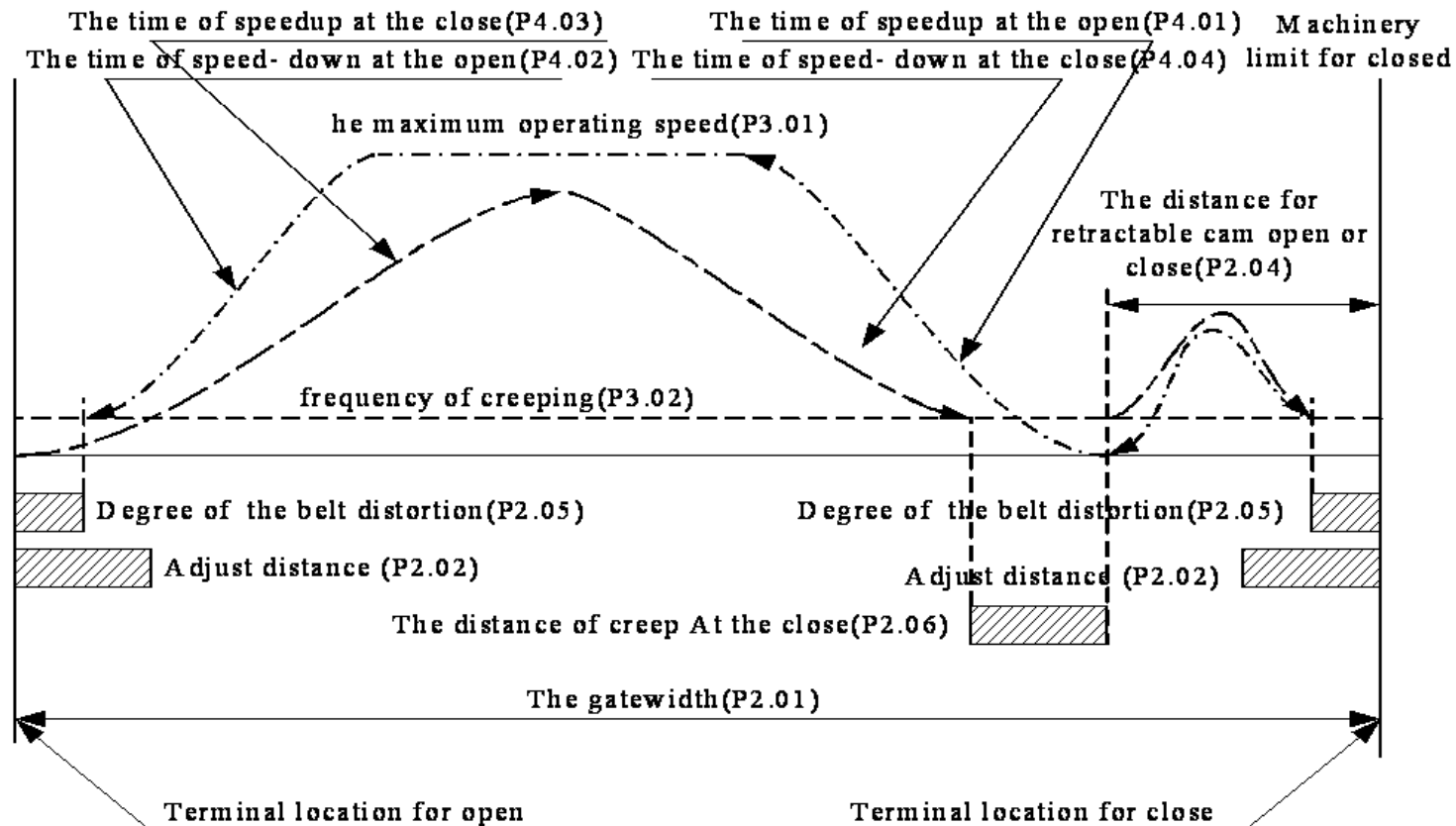


4.PM Motor Parameter

PM Motor Parameter			
Rated power	P	47	W
Rated voltage	Vn	125	V
Rated frequency	Fr	13	Hz
Rated Current	I	0.42	A
Rated speed	n	195	rpm
Rated torque	T	2.3	Nm
Locked Current	IO	1.45	A
Locked rotor torque	TO	8.1	Nm
RMS line-EMF	EMK	378	V/1000rpm
Poles		8	
Connection Type		Y	
20°C, Rw		150	Ω
Lw		589	m H
Protection class		IP44	
Insulation Class		F	

5. principle scheme for synchronous system

- - - - - The curve for opening (achieving the speed limit)
 — — — — The curve for closing (not achieving the speed limit)



6.Controller default parameter

Parameter NO.	Name	content	Gate machine Setting
P1.01.	Current Position	Just for the display	1
P1.02.	The numbers of Switching doors	The numbers of Switching doors,It plus one every Switching	0
P1.03.	Diameter of pulley (0.1mm)		22.4
P1.04.	reduction gear ratio		1.000
P1.05.	Type of the motor		3
P1.06	quick stopping	0: unavailability 1: validity	0
P1.07	Point barrier memory	0: unavailability 1: validity	1
P1.08	Demo mode	0: unavailability 1: validity	0
P1.09	Auto closing door at the first electrifying	0: unavailability 1: validity	0

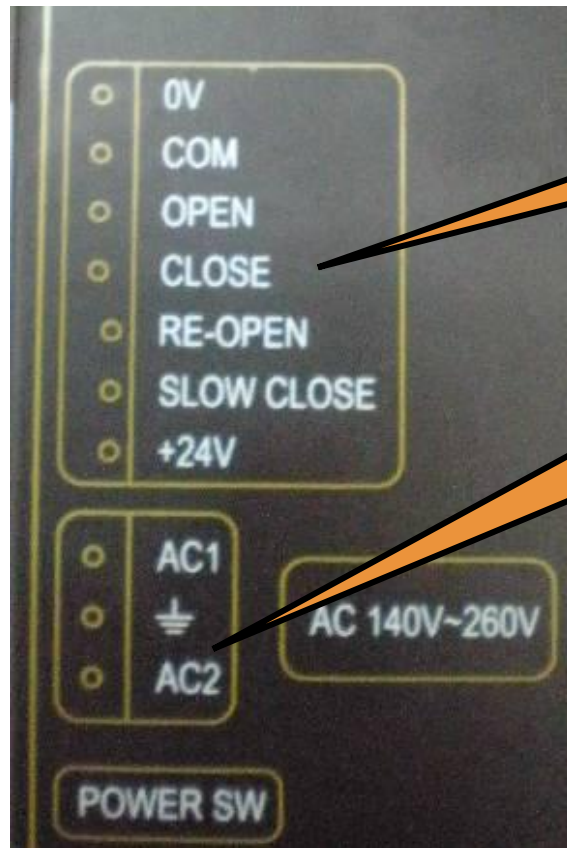
P2.01.	The gate width (mm)	The gate width from the Auto tuning of the driver	400
P2.02.	Adjust distance (0.1mm)	operating distance of Terminal signal	20
P2.03.	The distance of safety edge for door (mm)	The distance between the beginning of safety edge touching and the Terminal	90
P2.04	The distance for retractable cam open or close		45
P2.05	Degree of the belt distortion (0.1mm)	The distance between locked-rotor and the Terminal	43
P206	The distance of creep At the close(0.1mm)	The distance between emplacement At the close and the beginning of retracting or expanding retiring cam	0

P3.01.	The maximum operating speed(m/s)	The maximum permissible revolution of Servo System	0.60
P3.02.	frequency of creeping%	The Operating frequency at terminal area(The percentage rate of P3.01 as 100%)	5
P3.03.	frequency of Correction	The frequency of the first operation after electrifying for Correction location (The percentage rate of P3.01 as 100%)	30

Parameter NO.	Name	content	Gate machine Setting
P4.01.	The time of speedup at the open(S)		2.5
P4.02.	The time of speed-down at the open(S)		2.5
P4.03.	The time of speedup at the close(S)		2.5
P4.04.	The time of speed-down at the close(S)		2.5
P4.05.	The time of signal lag	The time of creeping after cancellation of signal of open or close	2.0
P4.06	The time of speedup and speed-down about retracting and putting out the retractable cam		1.5
P4.07	The time relay without input signal at	The time relay without input signal at the terminal of close	60
P4.08	The time of speed-down at middle		0.2
P4.09	The time of Torque with draw process		0.2

Parameter NO.	Name	content	Gate machine Setting
P5.01.	Torque-threshold of motion%	The maximum torque outer terminal Area	200.00
P5.02	Torque-threshold of keep- ing%	Torque-threshold of keeping and the maximum torque of the terminal	150.00
P5.03	Value of stall detection at close	Mesure the value of stalling torque for closing	180.0
P5.04.	The time for stall detection		0.3
P5.05.	The time for alignment(0.1S)	Alarm when the value exceeding the time for alignment	5.0

7.Controller Terminal definition



Input
Terminal

AC220V
Power Input



Output
Terminal



PM Motor port

Encoder port

Type	No.	No.	Signal Name	Function	Signal Level
AC220 Input	5	AC1	AC input L	AC input	AC 220V, +10%~-20% 50Hz/60Hz
	6	E	Grounding		
	7	AC2	AC input N		
Input signals	14	OV	GND	0V	24V~110V,AC/DC,15mA Photocoupler isolation
	13	COM	common		
	12	OPEN	Open signal	Termal prior levels from high to sl ow are Slow close signal, Open si gnal, Close signal	
	11	CLOSE	Close signal		
	10	RE-OPEN	Standby		
	9	SLOW-CLOSE	Slow close		
	8	+24V	+24V	+24V,80mA	
Output signa ls	17	MC	Common for output	Common for M1~M6	Dry contacts Contact cap acity AC250V DC30V Below 1A
	18	M1	Opened(N0)	Opened: M1-MC make contact, M 2-MC key out	
	19	M2	Opened(NC)		
	20	M3	Closed(NO)	Closed: M3-MC make contact, M4 -MC key out	
	21	M4	Closed(NC)		
	22	M5	Obstruction (N0)	Obstruction: M5-MC make contac t, M6-MC key out	
	23	M6	Obstruction (NC)		
	24	M7	Safety edge		
	25	M8			
MOTOR	27	U	U-phase phase	Power for the motor	3-phase 220VAC
	28	V	V-phase		
	29	W	W-phase		
	30	E	Grounding		

8.SW function



Switch	Definition
1	Door opening & do or closing direction
2	Test mode
3	Automatic learning
4	Active/non-active mode choice



1: change door opening direction

2: OFF-normal operation mode

3: OFF-learn motor magnetic pole mode

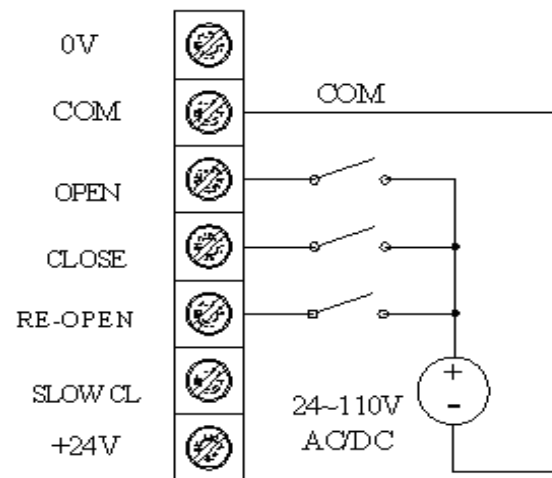
4: OFF-Non-active mode

ON- Automatic learning mode

ON-learn door width mode

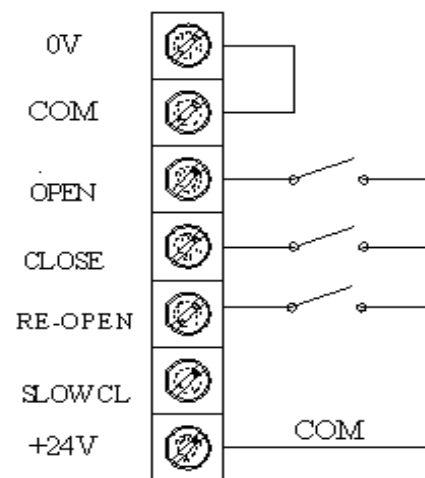
ON-active mode

9. Input signal wiring



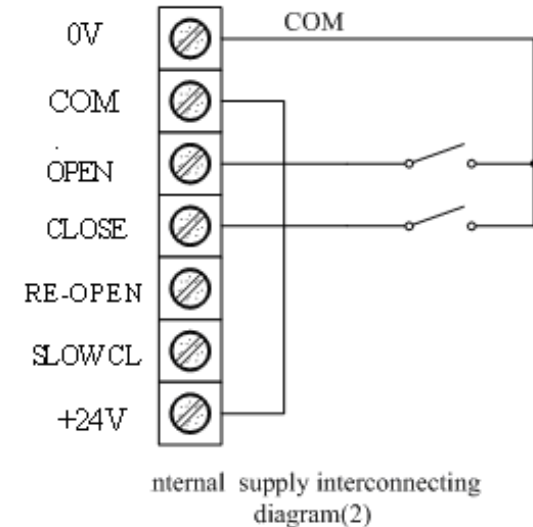
(Wiring at exterior power supply)

(Valid the voltage more than 18v)



(Wiring at interior power supply 1)

(valid of the contact closed)



internal supply interconnecting
diagram(2)



10. Running test method of door operator

(a) automatic learning of motor magnetic pole position

In the case without landing door, car is in non-opening terminal or closing terminal);

To select SW2-ON and others on "OFF" status, and press automatic learning button, and

wait 10s then automatic learning is completed.



Automatic
learning button

(b) door width automatic learning

Select SW

SW2-ON ,SW3-ON,



others on “OFF” status, and press automatic learning button. The motor will execute door opening and pause before door opening terminal and then close door until completely close the door.

r.

(c) Test running

Press test running button after successful automatic learning. The door will close and output door in place when the first press button. If the door open and output door in place, then the direction of running switch is wrong.)



When the position of the test running button is wrong, you can change it through SW1, i.e.: if SW1=ON, then pull SW1=OFF.

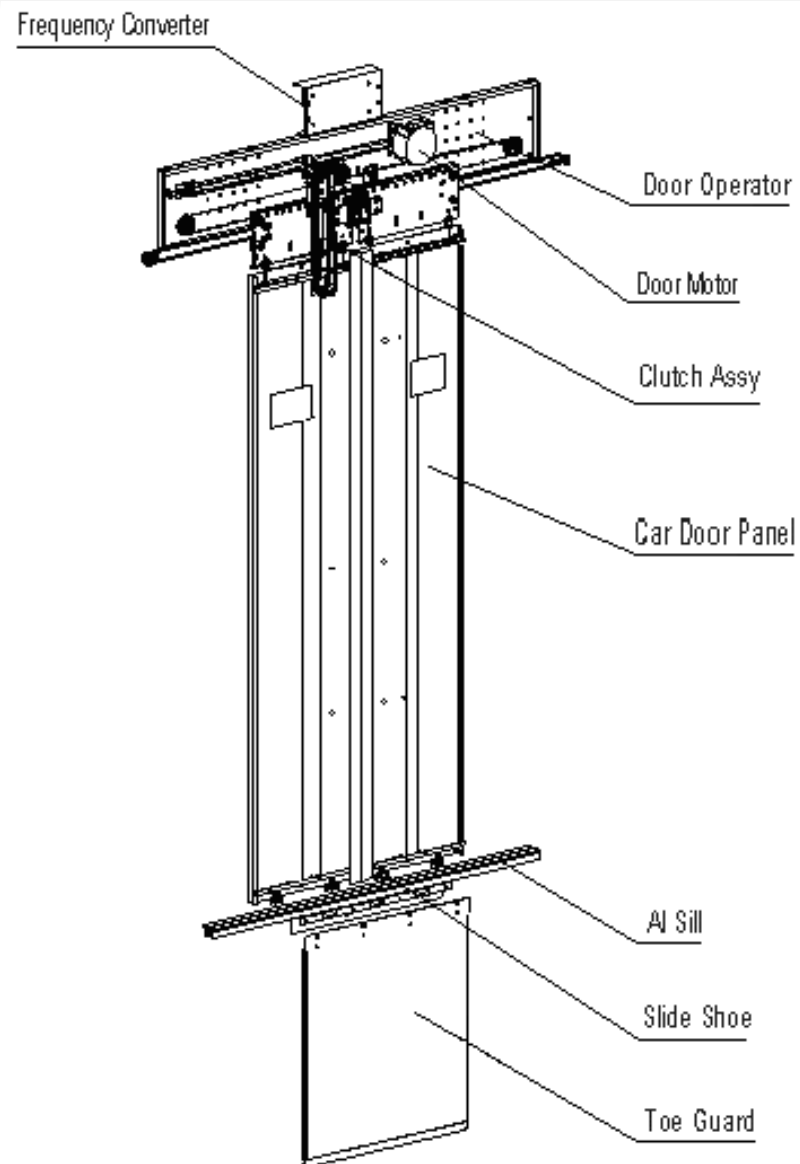
11. Door opening speed, keeping the speed of door opening & closing and keeping the force accommodation.

Directly adjust the door opening & closing speed of the motor through the knob, to maintain torque (refer to the following picture)

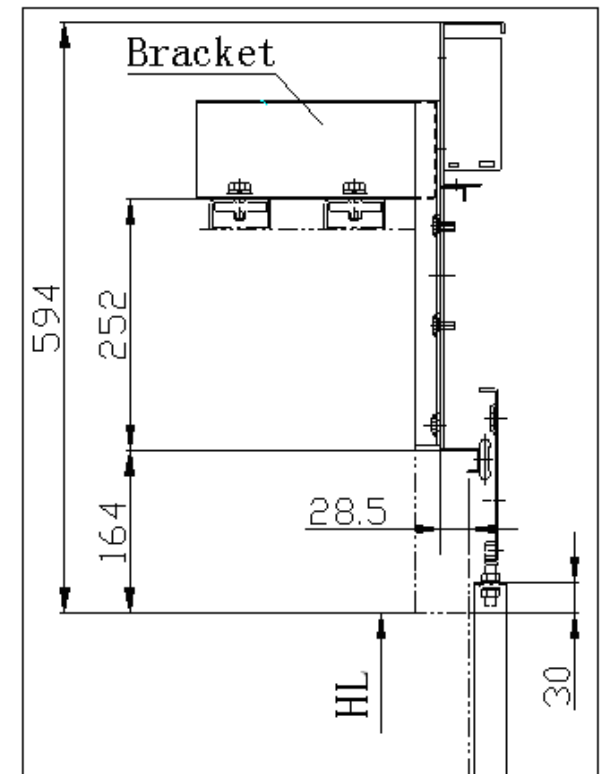
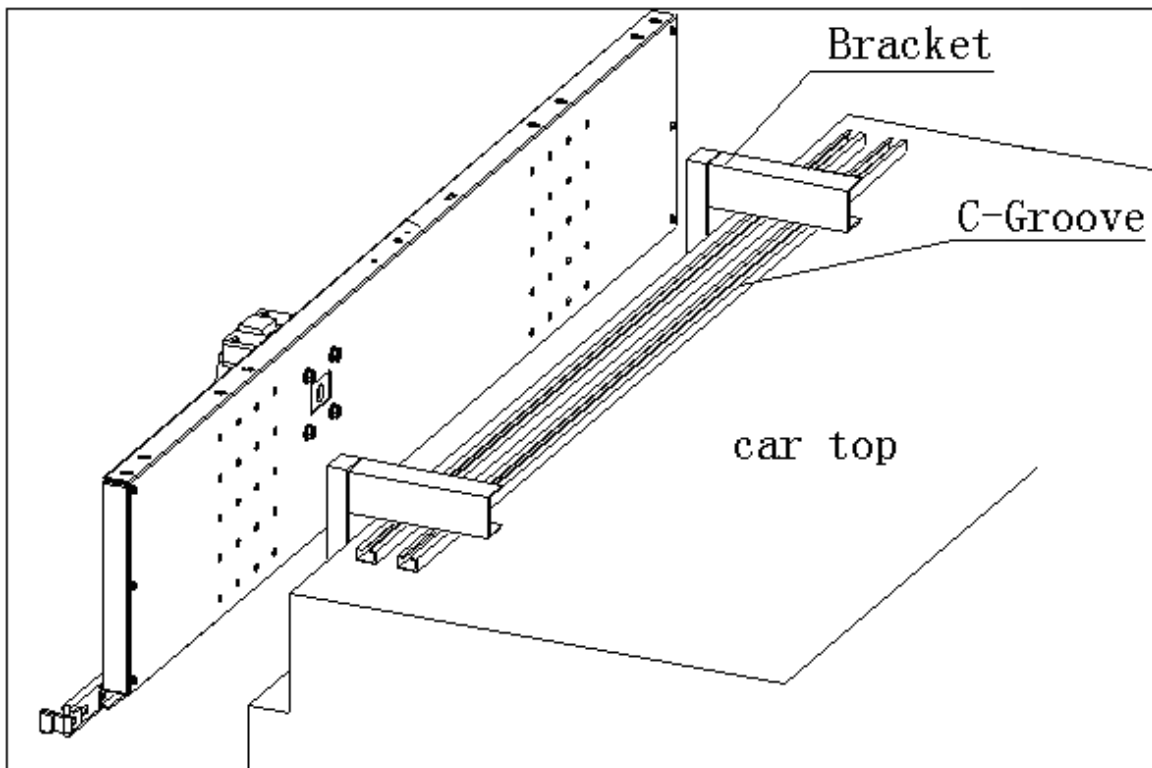


II .PM Door Sketch

1.PM Door Sketch



2.The connect between door motor module and elevator car



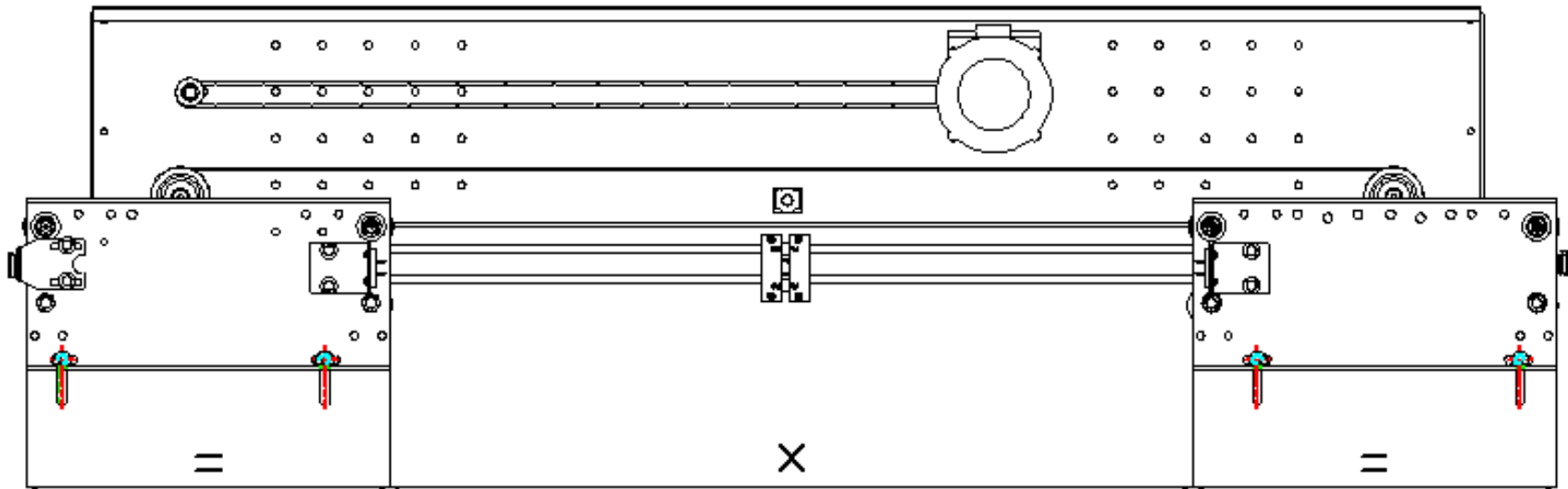
3. Adjustment of clear opening in centre parting doors

(Refer to the following picture.)

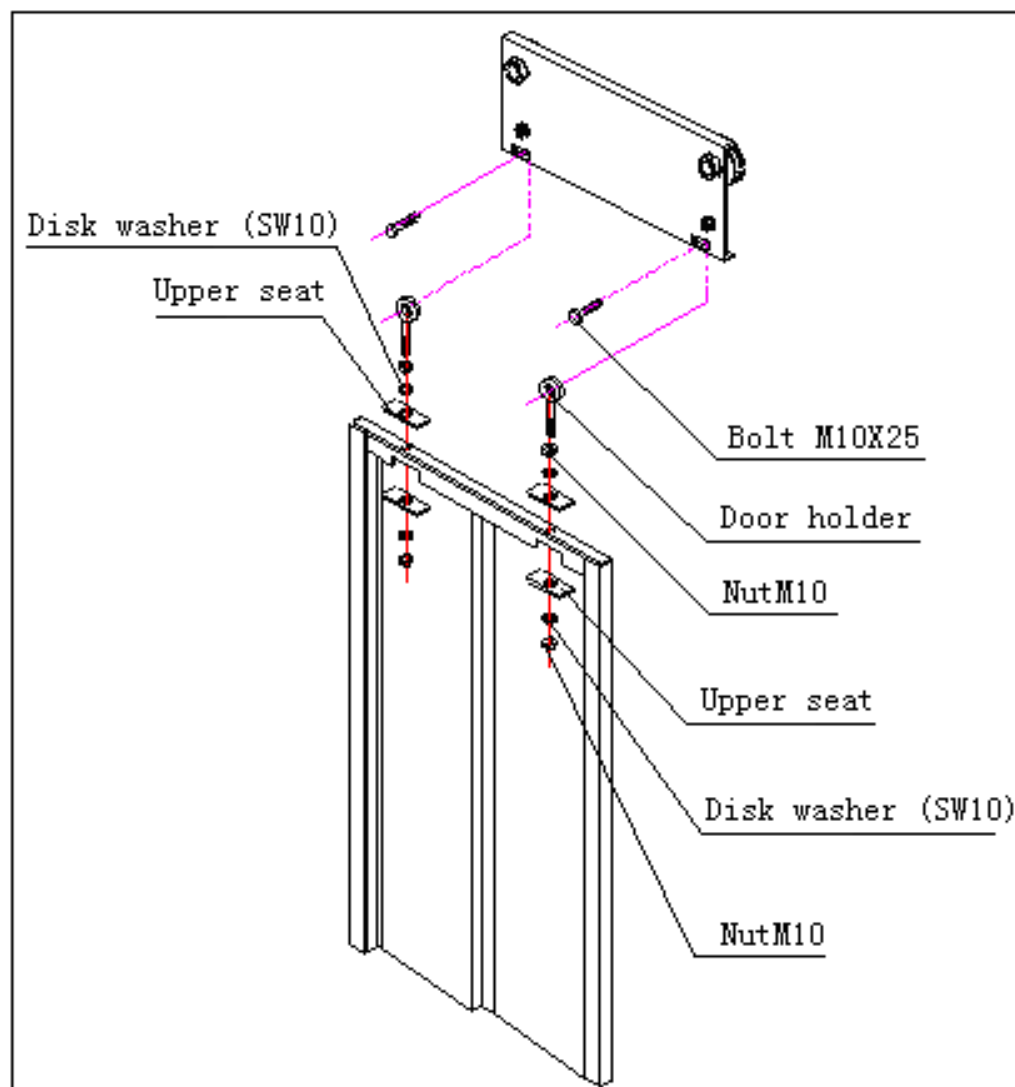
Instruction:

Place the flush plates in open position

Check its distance: $X = CO + 44$ (0, +2) = the distance between the plates when door opening.

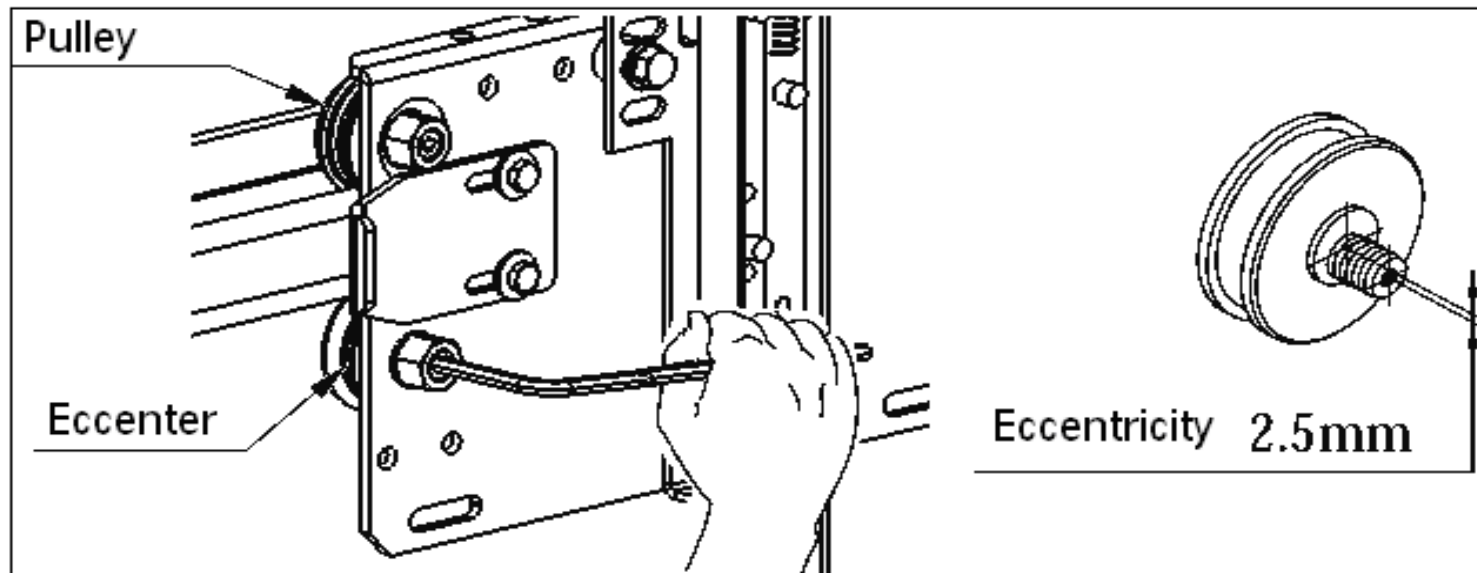


4. Installation of hangers



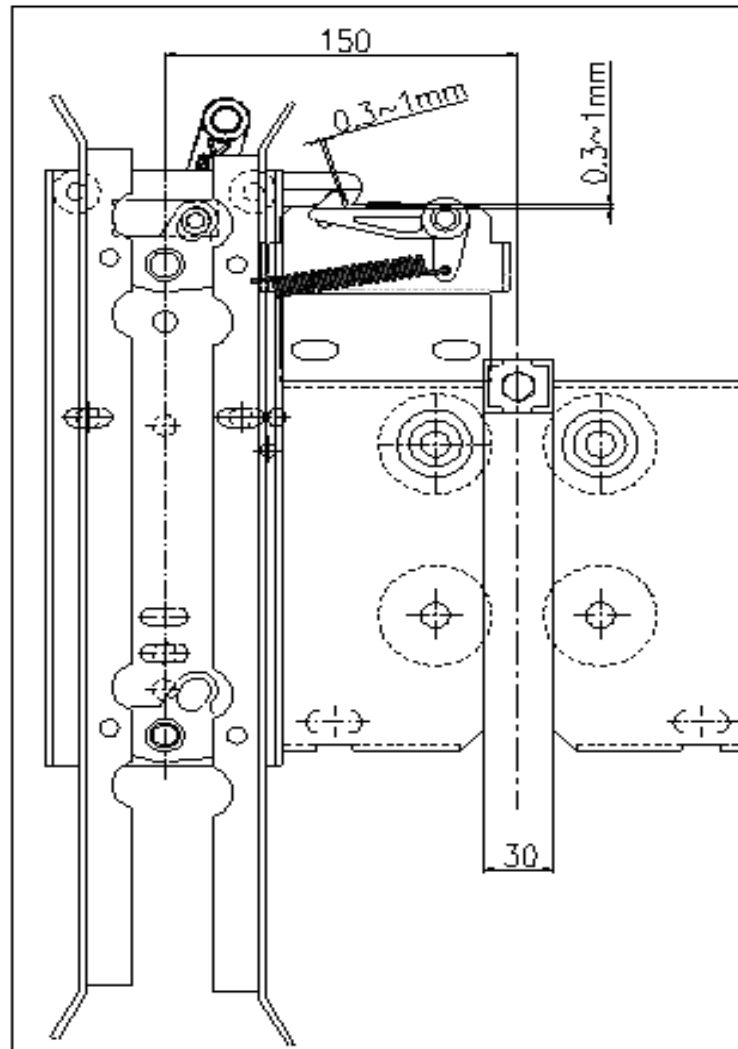
5.Adjustment of Eccenter

- The adjustment will be made by the eccentric, rotating the shaft we can adjust the force done to the track. See the following figure.
- This adjustment may be done with an Allen wrench, to make the distance in 0.2 ~0.5mm with the track, and then lock up the M10 nut. Pay attention that excessive tighten stop the panel.



*The eccentric was adjusted before out of the factory, and generally it does not need to adjust at site.

6. Once the skate is placed (135mm, between the end of the plate and shaft of the skate), in opened position; adjust the clamp-retainer set by the slots of the support with gap of 0.3~1mm.

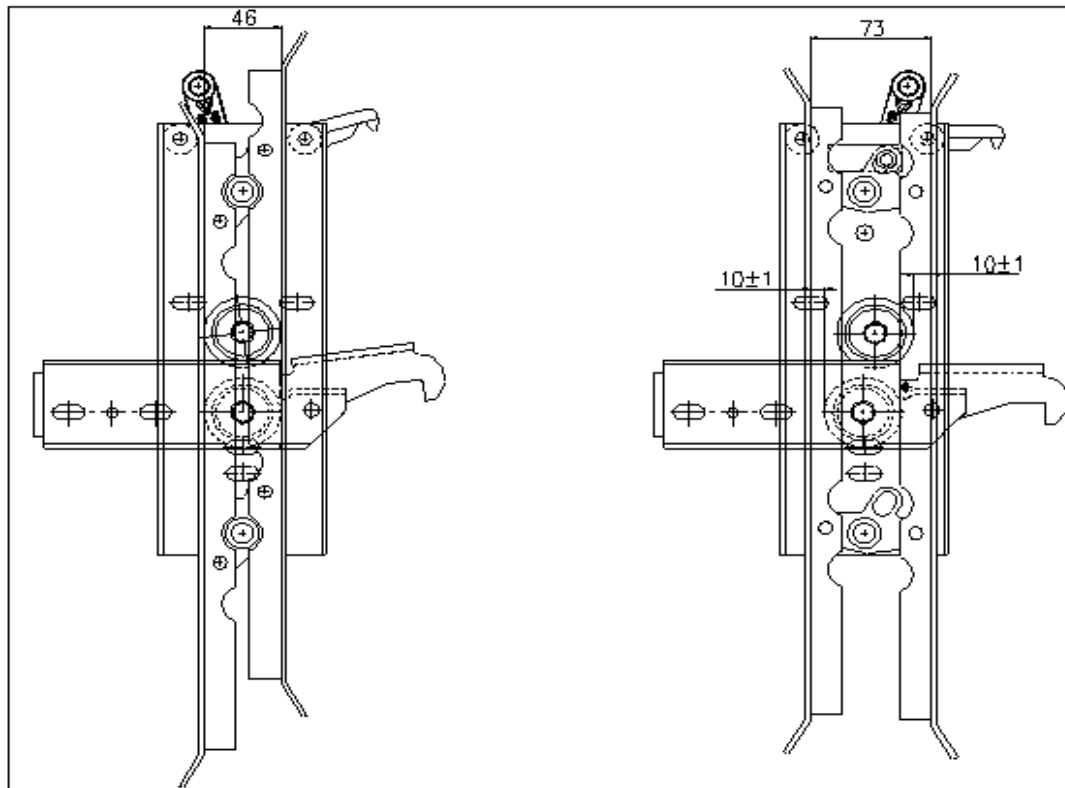


III. Fault Solution

1) The problem of the match of skate and unlock wheel cause door closing strike

Fault description and analysis: Elevator doors closing strike occurs in one landing. After checking, it is caused by the large gap between landing lock and the skate arm. The landing door closed earlier than the car door.

Solution: To check and adjust the distance to be 10 ± 1 mm between the skate arm and the upper landing lock and same distance between the skate arm and nether landing lock.



2) Too small gap of door panel cause crash

Fault description and analysis:

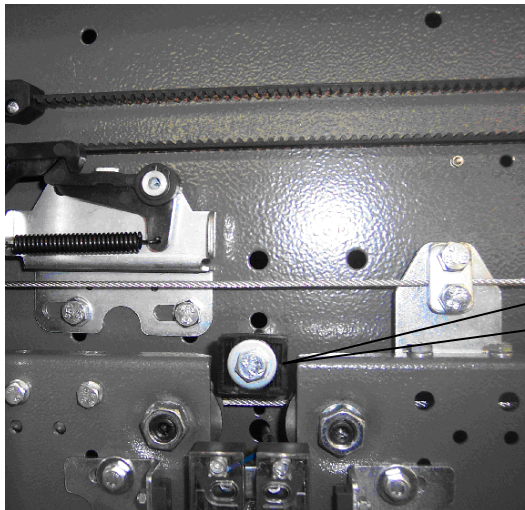
Crash happened when the door completely closed. After inspection, when the door completed closed, the rubber stopper did not contact with the plate and did not play the role of limit. There is no gap when the two doors shut and crash was caused when closing.

In some cases, door gap is not equal up and down, the top and bottom parts of the panel also had a slight collision when the door closing.

Solution:

To adjust again the distance to be 1~2 mm with the panel and be equal up and down after confirm the limit rubber block and the plated contact fully when the door complete closing, and then switch on the door inverter self-study function.

Prevention suggestion: to adjust the gap to be 1~2mm when installing the panel and to keep vertical.



No contact between the limit rubber block and the plate

3) Mechanical Resistance leads to not fully Door Opening.

Fault description and analysis:

Eccenter squeeze track, the running resistance of door is too large;

Too much resistance between guide shoe and tread;

Door slide block touches the bottom of tread, panel touches tread;

Too big tension of compulsion door closing spring;

Steel wire rope, teeth-shape belt tightening, increased running resistance

Solution

a) To adjust the gap to be 0.2-0.5mm between eccentric and track;

b) To clean tread, correct panel, adjust slide block, make to run smoothly;

c) To adjust the gap to be 3-5mm between panel and tread.

d) To change the position of spring hook, to reduce the tension.

e) To adjust the tightness of steel wire rope and teeth-shape belt.

f) To ask the service if it can not solve the problem after the above improvement.

Prevention suggestion :

a) To read the detailed instruction in manual during installation.

b) To check the motor running resistance when installation.

4) Door opening crash as skate is not fully open

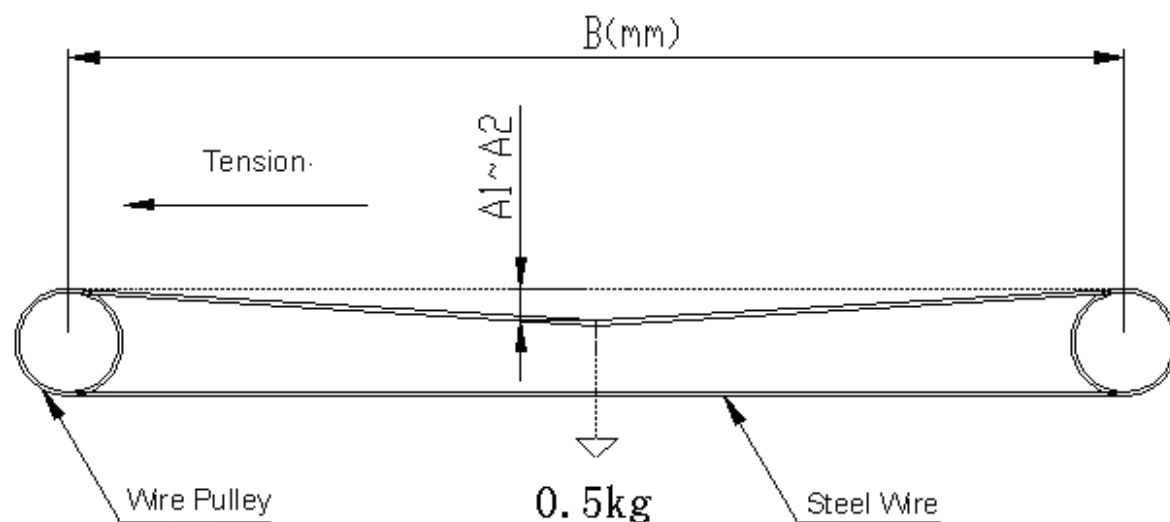
Fault description and analysis: Open the door, the skate is not fully contracted when opening, the process did not crawl through the low speed when close to the door stopper block. Then the crash was happened when the door was fully open .When door open and close, car door and landing door are fully synchronized. In the process of door opening, the skate first fully contracted and opened the landing door lock at the same time before the plates operated driven by the skate, and then drag by the tooth belt to run car door and landing door. Toothed belt drive the skate full contraction, requires 7mm moving range.

In this case, as the skate is not completely contracted and the moving range of the teeth-shape belt should be 7mm to stretch the skate, the belt is still pulling the door running at the high speed when the door open to the limited position. This leads to door opening crash.

Solution:

To check and adjust the resistance of the skate itself. And to check the interaction between skate and lock open wheel, adjust the distance to be 10mm between the front wheel and skate arm, rear wheel and the arm.

5) To check every time the door rope, belt tension level during regular maintenance



PL	B	A1	A2
700	815	14	18
800	865	14	18
900	915	14	18
1000	965	14	18
1100	1015	14	18
1200	1100	14	18
1300	1150	14	18

Thank you !